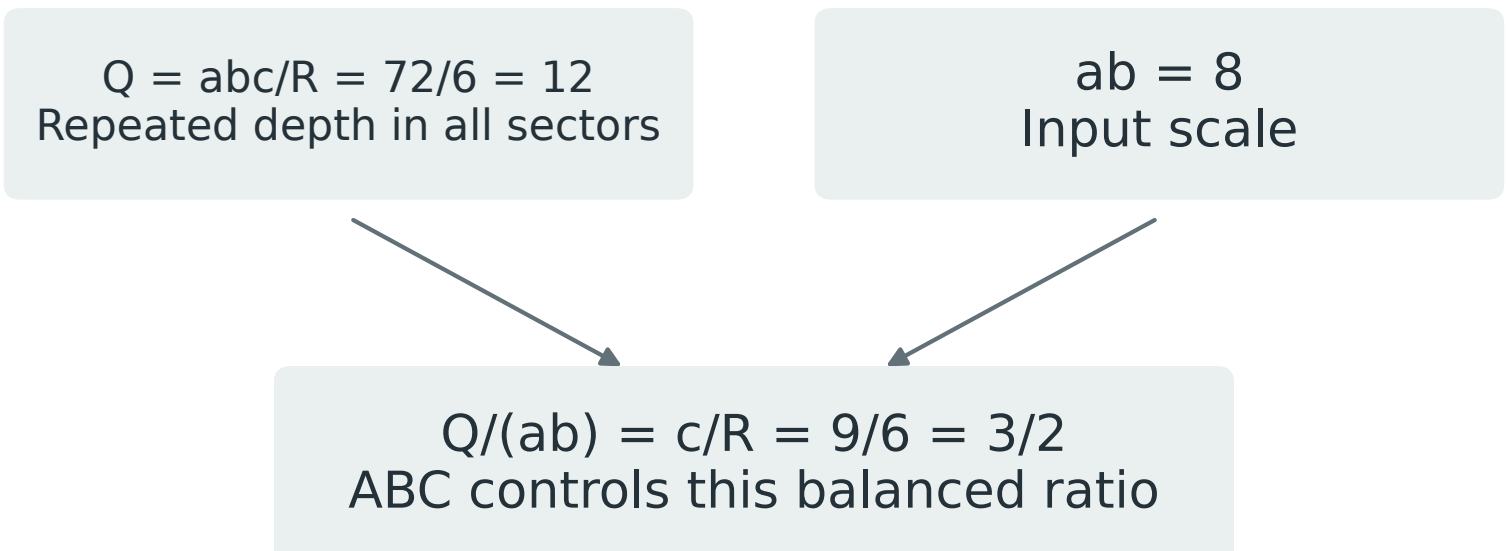
Keep the input factor ab in the normalization



Q alone cannot be uniformly subpolynomial in R.